Circles and Polygons — Radius/Diameter and Area/Perimeter Errors
Answer Key

Grades 6–7

Quick Answers

1. 31.42	3. 113.10	5. 52	7. 154.06
2. 153.94	4. 26	6. 5.65	8. —

Common wrong answers

1. *If they got 15.71: used C = pi times r, which is the circumference formula with the radius dropped into the slot that wants the diameter*
2. *If they got 615.75: substituted the diameter straight into A = pi r squared without halving it first*
3. *If they got 37.70: used the circumference formula 2 pi r for a question that asked for the area*
4. *If they got 36: multiplied the two side lengths, which is the area of the garden rather than the edging around it*
5. *If they got 34: added the six side lengths, which measures the fence around the plot instead of the ground it covers*
6. *If they got 11.31: used 2 pi times the measurement given, treating a diameter as if it were a radius*
7. *If they got 6082.12: dropped the given circumference straight into A = pi r squared as though it were the radius*

1. Circumference of a badge with radius 5 cm.

Step 1: the question asks for the distance around the circle, so the tool is $C = 2\pi r$ — and the given measurement is already a radius, so nothing needs halving or doubling.
Step 2: substitute: $C = 2\pi(5) = 10\pi$.
Step 3: $10\pi = 31.4159\ldots$, so $C \approx \boxed{31.42 \text{ cm}}$.
Watch for: 15.71. That is πr — the πd form with a radius pushed into the diameter's slot. It is exactly half the right answer, which is the signature of this swap.

2. Area of a tabletop whose diameter is 14 cm.

Step 1: the question asks how much surface the top covers, so the tool is $A = \pi r^2$. That formula takes a *radius*, and 14 cm is a diameter, so convert before substituting.
Step 2: $r = 14 \div 2 = 7$ cm.
Step 3: $A = \pi(7)^2 = 49\pi = 153.938\ldots$, so $A \approx \boxed{153.94 \text{ cm}^2}$.
Watch for: 615.75. That is $\pi(14)^2$ — the diameter squared. Because the radius is squared, forgetting to halve multiplies the answer by four, not by two.

3. Find and fix: Rae's circle of radius 6 cm.

Step 1: name what Rae used. $2\pi r$ is the **circumference** formula, so her 37.70 is a length in cm — the distance once around the circle. Her own unit label, cm^2, contradicts the formula she wrote, which is the clue.
Step 2: the question asked for area, so use $A = \pi r^2$ with the radius she was already given.
Step 3: $A = \pi(6)^2 = 36\pi = 113.097\ldots$, so $A \approx \boxed{113.10 \text{ cm}^2}$.
Full-credit answer says: "Rae used the circumference formula, which measures the distance around, not the surface covered," and then shows πr^2.

4. Find and fix: Marco's 9 m by 4 m garden bed.

Step 1: name what Marco computed. Length times width is the **area** of the rectangle, so his 36 is a number of square metres of soil — not a length of edging.
Step 2: edging runs around the outside, so the question wants the perimeter: $P = 2(9) + 2(4) = 18 + 8$.
Step 3: the bed needs $\boxed{26 \text{ m}}$ of edging.
Full-credit answer says: "Marco found the area instead of the perimeter." Note that his mistake would have cost him 10 extra metres of edging, and that a length can never be measured in square metres.

5. Area of the L-shaped plot $(0,0),(10,0),(10,4),(4,4),(4,7),(0,7)$.

Step 1: the landscaper wants the ground covered, so this is area. The plot is not a single rectangle, so split it into two rectangles rather than reaching for one formula.
Step 2: cut along the line $y = 4$. The bottom piece runs from $x = 0$ to $x = 10$ and from $y = 0$ to $y = 4$: area $10 \times 4 = 40$. The top piece runs from $x = 0$ to $x = 4$ and from $y = 4$ to $y = 7$: area $4 \times 3 = 12$.
Step 3: add the pieces: $40 + 12$, giving $\boxed{52 \text{ square units}}$.
Step 4: check another way — the full 10 by 7 rectangle is 70, minus the missing 6 by 3 corner, which is 18: $70 - 18 = 52$ ✓.
Watch for: 34. That is the sum of the six side lengths, i.e. the perimeter of the plot. A fence and a lawn are different orders (metres versus square metres).

6. Trim around a rug measuring 1.8 m edge-to-edge through the centre.

Step 1: classify the given number first. A measurement taken *through the centre* from edge to edge is a **diameter**, not a radius.
Step 2: trim runs around the rug, so use the circumference. With a diameter in hand the direct form is $C = \pi d$, no halving and no doubling: $C = \pi(1.8)$.
Step 3: $\pi(1.8) = 5.6548\ldots$, so $C \approx \boxed{5.65 \text{ m}}$.
Watch for: 11.31. That is $2\pi(1.8)$ — the $2\pi r$ form fed a diameter, which doubles the answer.

7. Area of a pool whose railing is 44 m around.

Step 1: the given number is a length around the circle, so it is the circumference. The area formula needs a radius, so recover r from C before doing anything else.
Step 2: from $C = 2\pi r$, $r = \dfrac{44}{2\pi} = 7.0028\ldots$ m.
Step 3: $A = \pi r^2 = \pi\left(\dfrac{44}{2\pi}\right)^2 = 154.0619\ldots$, so $A \approx \boxed{154.06 \text{ m}^2}$.
Step 4: sanity check — a radius of about 7 m gives an area near $\pi(7)^2 \approx 154$ ✓.
Watch for: 6082.12. That is $\pi(44)^2$: the circumference dropped straight into the radius slot. An answer forty times too large is the tell.

8. Explain: circle B has twice the radius of circle A.

Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.

(a) Write both quantities for a radius of r and then for $2r$. Circumference: $2\pi r$ becomes $2\pi(2r) = 2\times(2\pi r)$ — the radius appears to the first power, so doubling it doubles the length. Area: πr^2 becomes $\pi(2r)^2 = \pi \cdot 4r^2 = 4 \times (\pi r^2)$ — the radius is squared, so the scale factor is squared too. Doubling a radius doubles a length and quadruples a covering.

(b) Model habit: "*First* ask whether the question wants a trip around the shape (length: use $2\pi r$ or add the sides) or the surface it covers (covering: use πr^2 or multiply). *Then* check that my answer's unit matches — m for a length, m^2 for a covering."

(c) Rae wrote a circumference formula but labelled her answer cm^2. Part two of the habit compares the formula with the unit, and those two disagree, so the check fires before the answer is ever written down.

What a full-credit answer must contain: the substitution $r \to 2r$ in both formulas, the observation that squaring the radius squares the scale factor, a habit with both a decide-first part and a unit-check part, and an explicit link back to the mismatch in problem 3.